Let $a \in S$. It is to be proved that the inverse a (if exists) is unique.

Suppose that a has two inverses, say, a_1, a_2.

Treating a_1 as an inverse of a gives $a * a_1 = a_1 * a = e$ …(1)

Next treating a_2 as the inverse of a gives $a * a_2 = a_2 * a = e$ …(2)

$$a_1 = a_1 * e = a_1 * (a * a_2) = (a_1 * a) * a_2 = e * a_2 = a_2 \text{ (by (1) and (2)).}$$

So, $a_1 = a_2$. Hence the inverse of a is unique which completes the proof.

Example 12.2

Verify the (i) closure property, (ii) commutative property, (iii) associative property (iv) existence of identity and (v) existence of inverse for the arithmetic operation $+$ on $\mathbb{Z}$.

Solution

(i) $m+n \in \mathbb{Z}, \forall m, n \in \mathbb{Z}$. Hence $+$ is a binary operation on $\mathbb{Z}$.

(ii) Also $m+n = n+m, \forall m, n \in \mathbb{Z}$. So the commutative property is satisfied

(iii) $\forall m, n, p \in \mathbb{Z}, m+(n+p) = (m+n)+p$. Hence the associative property is satisfied.

(iv) $m+e = e+m = m \Rightarrow e = 0$. Thus $\exists\ 0 \in \mathbb{Z} \ni (m+0) = (0+m) = m$. Hence the existence of identity is assured.

(v) $m+m' = m'+m = 0 \Rightarrow m' = -m$. Thus $\forall m \in \mathbb{Z}, \exists\ -m \in \mathbb{Z} \ni$ $m+(-m) = (-m)+m = 0$. Hence, the existence of inverse property is also assured. Thus we see that the usual addition $+$ on $\mathbb{Z}$ satisfies all the above five properties.

Note that the **additive identity** is 0 and the **additive inverse** of any integer m is $-m$.

Example 12.3

Verify the (i) closure property, (ii) commutative property, (iii) associative property (iv) existence of identity and (v) existence of inverse for the arithmetic operation $-$ on $\mathbb{Z}$.

Solution

(i) Though $-$ is not binary on $\mathbb{N}$; it is binary on $\mathbb{Z}$. To check the validity of any more properties satisfied by – on $\mathbb{Z}$, it is better to check them for some particular simple values.

(ii) Take $m = 4$, $n = 5$ and $(m-n) = (4-5) = -1$ and $(n-m) = (5-4) = 1$.

Hence $(m-n) \neq (n-m)$. So the operation $-$ is not commutative on $\mathbb{Z}$.

(iii) In order to check the associative property, let us put $m = 4, n = 5$ and $p = 7$ in both $(m-n)-p$ and $m-(n-p)$.

$(m-n)-p = (4-5)-7 = (-1-7) = -8$ …(1)

$m-(n-p) = 4-(5-7) = (4+2) = 6$. …(2)

From (1) and (2), it follows that $(m-n)-p \neq m-(n-p)$.

Hence – is not associative on $\mathbb{Z}$.

(iv) Identity does not exist (why?).

(v) Inverse does not exist (why?).

Example 12.4

Verify the (i) closure property, (ii) commutative property, (iii) associative property (iv) existence of identity and (v) existence of inverse for the arithmetic operation $+$ on $\mathbb{Z}_e$ = the set of all even integers.

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